THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · NCERT-TYPE UNIT TEST SET 3

Chapters 9–12: **Light - Reflection and Refraction, The Human Eye and the Colourful World, Electricity, Magnetic Effects of Electric Current** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The V-I graph for an ohmic conductor is 1

(a) a straight line through the origin **(b)** a parabola
(c) a horizontal line **(d)** a curve

2. An electromagnet is made of 1

(a) plastic
(b) copper only
(c) soft iron wrapped with a coil carrying current
(d) steel only

3. SI unit of power of a lens is 1

(a) dioptre **(b)** newton **(c)** metre **(d)** watt

4. Which colour bends most on passing through a prism? 1

(a) Red **(b)** Violet **(c)** Yellow **(d)** Orange

5. In a parallel circuit the quantity that is the same across all the resistors is 1

(a) resistance **(b)** current
(c) power **(d)** potential difference

6. An induced current is produced in a coil when 1

(a) the coil is kept in a steady field at rest
(b) a magnet is moved towards or away from it
(c) a magnet is held still near it
(d) a battery is disconnected

7. The focal length of a spherical mirror of radius of curvature R is 1

(a) R **(b)** R/4 **(c)** R/2 **(d)** 2R

8. The ability of the eye lens to adjust its focal length is called 1

(a) myopia **(b)** dispersion **(c)** scattering **(d)** accommodation

9. Potential difference is measured by a 1

(a) ammeter in parallel **(b)** voltmeter connected in series
(c) galvanometer in series **(d)** voltmeter connected in parallel

10. The strength of a magnetic field is indicated by 1

(a) the length of the lines **(b)** the number of poles

(c) the closeness of the field lines **(d)** the colour of the lines

11. If light enters a medium of refractive index 1.5 from vacuum, its speed becomes (c = 3×10^8 m/s) 1

(a) 2×10^8 m/s **(b)** 1.5×10^8 m/s **(c)** 4.5×10^8 m/s **(d)** 3×10^8 m/s

12. In hypermetropia, the near point is 1

(a) nearer than 25 cm **(b)** at 25 cm

(c) at infinity **(d)** farther than 25 cm

13. The resistance of a conductor is 1

(a) inversely proportional to its length **(b)** directly proportional to its length

(c) independent of length **(d)** proportional to the square of the area

14. The strength of an electromagnet can be increased by 1

(a) decreasing the current

(b) removing the core

(c) increasing the current or the number of turns

(d) using a thicker iron-free coil

15. The power of a concave lens is 1

(a) negative **(b)** positive **(c)** zero **(d)** infinite

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Write the expressions for power in three forms. 2

17. State two advantages of AC over DC. 2

18. Define the principal focus of a concave mirror. 2

19. Name the part of the eye which controls the amount of light entering and its working. 2

20. What is meant by the electric potential difference between two points? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. How much work is done in moving a charge of 2 C across two points having a potential difference of 12 V? 3

22. Describe Oersted's experiment and state the conclusion. 3

23. An object is placed 20 cm from a concave lens of focal length 20 cm. Find the image distance. 3

24. What is a cataract? How is it treated? 3

OR

Three resistors 20 ohm, 30 ohm and 60 ohm are in parallel. Find the equivalent resistance. 3

25. State Fleming's left-hand rule with the help of the three fingers. 3

OR

What does the sign of magnification tell about the image? Give the sign for real and virtual images. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. State the mirror formula and the sign convention. An object 4 cm tall is placed 25 cm from a concave mirror of focal length 20 cm. Find the image position, size and nature. 5

OR

What is accommodation? Explain how the eye focuses on objects at different distances and mention the near and far points. 5

27. Derive the formula for the equivalent resistance of three resistors connected in series. Find it for 3 ohm, 4 ohm and 5 ohm. 5

OR

Describe the magnetic field due to a current-carrying solenoid with the field pattern. Explain how it is used to make an electromagnet. 5

END OF QUESTION PAPER Total: 50 Marks

MASTERMIND